

Nomthi MHLANGU

Exercise 1 - SQL FOUNDATIONS

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SELECTING & SORTING

employees						
id	first-name	last-name	department	salary	hire-date	city
1.	Alice	Green	IT	70000	2020-01-10	Johannesburg
2.	Brian	Lee	HR	45 000	2019-03-22	Cape Town
3.	Cathy	Zulu	Finance	65 000	2018-07-18	Durban
4.	David	Mokoena	Marketing	50 000	2021-11-05	Pretoria
5.	Eva	Naidoo	IT	72 000	2017-09-30	Johannesburg

1. Retrieve all columns from the employees table

```
SELECT *  
FROM employees;
```

id	first-name	last-name	department	salary	hire-date	city
1	Alice	Green	IT	70000	2020-01-10	Johannesburg
2	Brian	Lee	HR	45 000	2019-03-22	Cape town
3.	Cathy	Zulu	Finance	65 000	2018-07-18	Durban
4.	David	Mokoena	Marketing	50 000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72 000	2017-09-30	Johannesburg

2. Find all unique departments.

```
SELECT DISTINCT department  
FROM employees;
```

department
IT
HR
Finance
Marketing

3. Retrieve first and last names ordered by salary descending

```
SELECT first-name, last-name  
FROM employees  
ORDER BY salary DESC;
```

first-name	last-name
Eva	Naidoo
Alice	Green
Cathy	Zulu
David	Mokwena
Brian	Lee

4. Retrieve the top 3 highest-paid employees

```
SELECT id, first-name, last-name, salary  
FROM employees  
ORDER BY salary DESC  
LIMIT 3;
```

id	first-name	last-name	salary
5	Eva	Naidoo	72 000
1	Alice	Green	70 000
3	Cathy	Zulu	65 000

Section: 2 Filtering with WHERE

5. Find employees in the IT department

```
SELECT id, first-name, last-name, department  
FROM employees  
WHERE department = 'IT';
```

id	first-name	last-name	department
1	Alice	Green	IT
5	Eva	Naidoo	IT

6. Find employees in Finance with salary > 60 000

```
SELECT id, first-name, last-name, department, salary  
FROM employees  
WHERE department = 'Finance' AND salary > 60 000;
```

id	first-name	last-name	department	salary
3	Cathy	Zulu	Finance	65 000

7. Find employees in HR or Marketing.

```
SELECT id, first-name, last-name, department  
FROM employees  
WHERE department = 'HR' OR department = 'Marketing';
```

id	first-name	last-name	department
2	Brian	Lee	HR
4	David	Mokoena	Marketing

8. Find employees not in IT

```
SELECT id, first-name, last-name, department
```

id	first-name	last-name	department
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Mokoena	Marketing

9. Find employees in IT, HR or Finance using IN

```
SELECT id, first-name, last-name, department  
FROM employees  
WHERE department IN ('IT', 'HR', 'Finance');
```


id	first_name	last_name	department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
5	Eva	Naidoo	IT

10. Find employees in IT with salary > 65 000 and city Johannesburg

SELECT id, first_name, last_name, department, salary, city
FROM employees

WHERE department = 'IT' AND salary > 65 000 AND city = 'Johannesburg';

id	first_name	last_name	department	salary	city
1	Alice	Green	IT	70 000	Johannesburg
5	Eva	Naidoo	IT	72 000	Johannesburg

END